

Answer key and marking scheme - not for distribution.

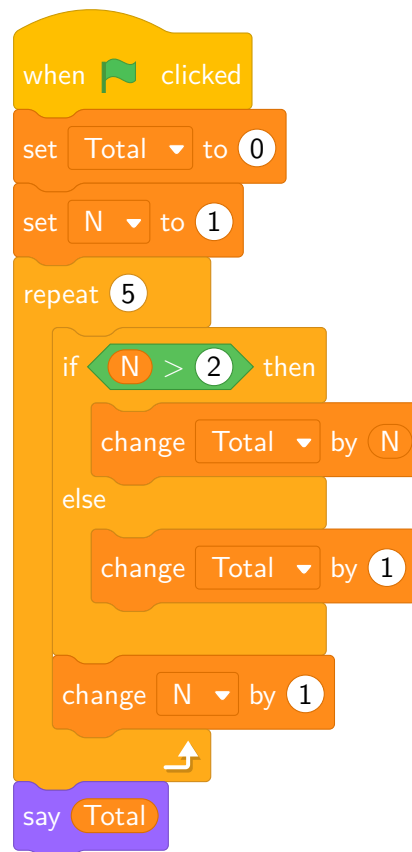
Time: 10 minutes

Maximum Marks: 4

Question 1 - Predict the output

[4 marks]

The **Total** and **N** variables have already been created. Work through the script below **one block at a time** and answer the questions under it. Do not run it in Scratch - the point is to predict what it will do. Keeping a rough trace table in the margin is the easiest way to track the variables.



1 (a) The output

[2 marks]

What number does the script **say** when it finishes?

Answer.

It says **14**.

2 marks for 14. Award 1 mark for a correct method with one arithmetic slip. The common wrong answer is 15 - that is what you get by adding every N and ignoring the else branch.

1 (b) The counter

[1 mark]

What is the value of **N** when the script finishes?

Answer.

6.

The usual wrong answer is 5. `change N by 1` is the last block inside the loop, so it also runs on the final pass - N is left one past the last value it was tested with.

1 (c) The else branch

[1 mark]

How many times does the **else** branch run?

Answer.

2 times (on the passes where N is 1 and 2).

The full trace:

<i>Pass</i>	<i>N at start</i>	<i>Is $N > 2$?</i>	<i>Total after</i>
1	1	No	1
2	2	No	2
3	3	Yes	5
4	4	Yes	9
5	5	Yes	14

After the last pass `change N by 1` runs once more, so N finishes at 6, not 5.

- End of Question Paper -